Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques.

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1. Introduction

Radiation dosimetry is crucial for monitoring absorbed doses across various fields. Biomass from sustainable materials, such as cowrie shells, have shown promise for this purpose. Primarily composed of calcium carbonate, they exhibit a luminescent response to high radiation doses, positioning them as a viable alternative for high dose dosimetry (Ogundare et al., 2021).

2. Materials and Methods

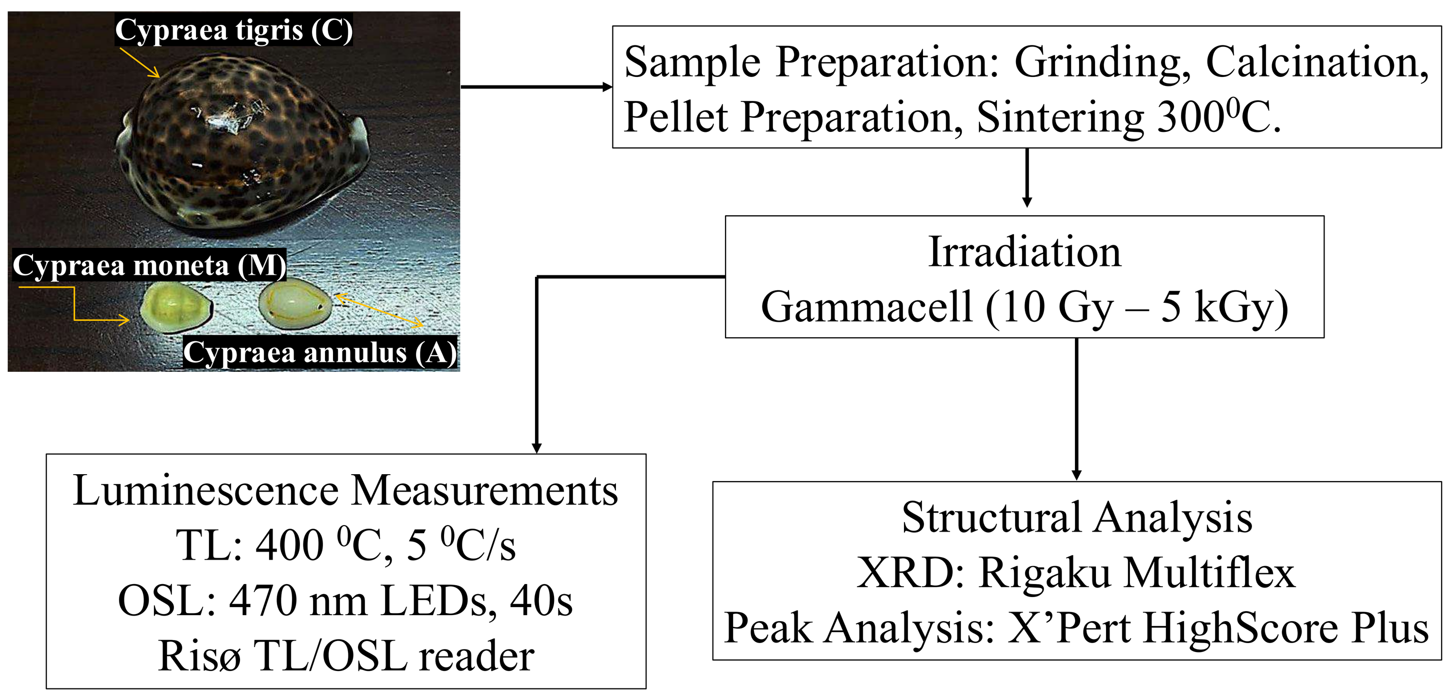

Figure 1: Experimental workflow for cowrie shells dosimetric characterization

Figure 1 presents a comprehensive overview of the dosimetric pellet production workflow, from the fabrication steps to the analytical methods and software chosen for accurate results interpretation.

3. Results and Discussion

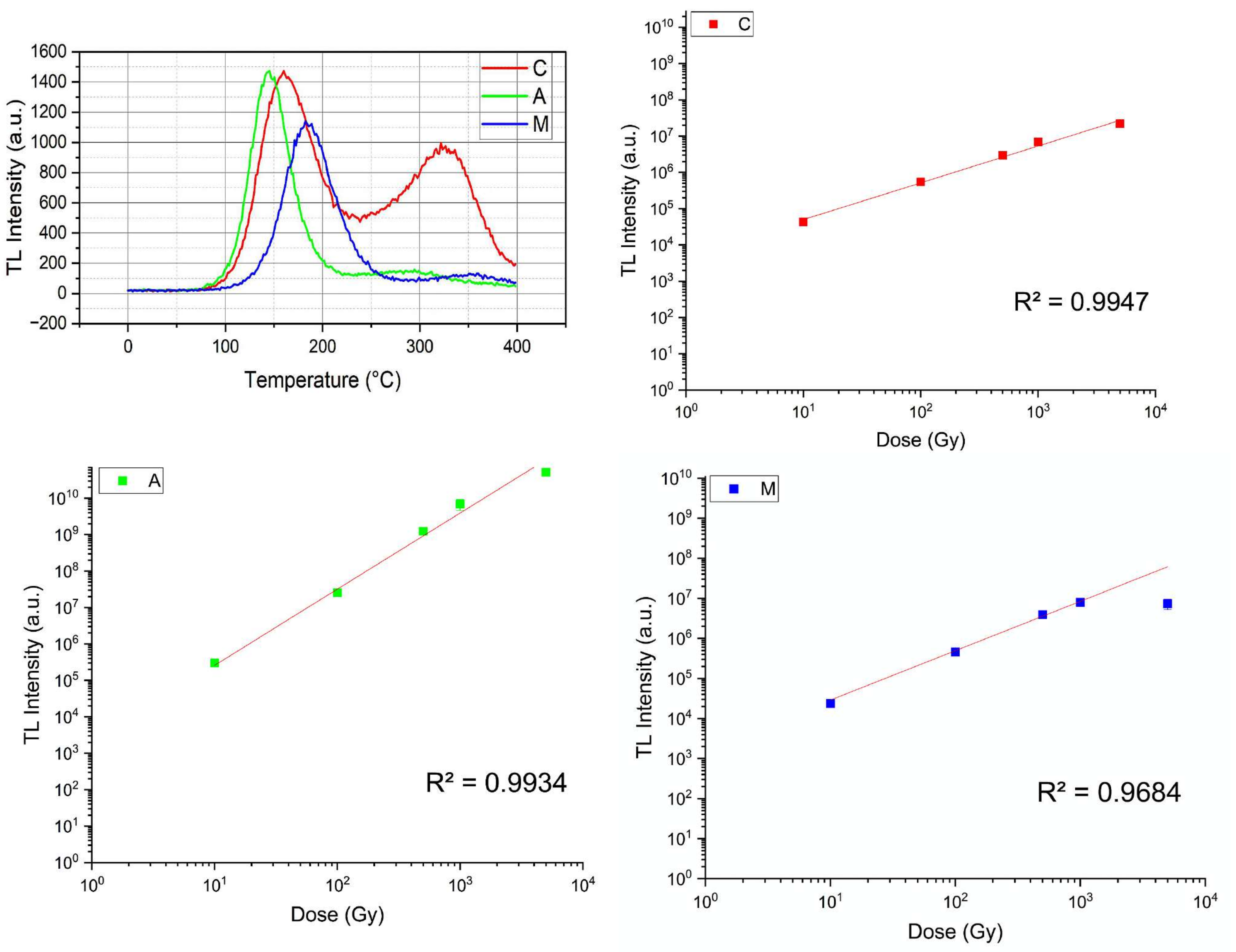

Figure 2: TL Glow curves and dose–response curves of the samples C, A and M shells in the range of 10 Gy – 5 kGy.

The cowrie shell luminescence analyses showed a linear dose response ($R^2 > 0.99$). TL exhibited prominent peaks between 150 °C and 170 °C, with sample A showing higher intensity at lower temperatures (Figure 2), while sample C displayed a broader glow curve. In OSL, sample C presented the highest sensitivity of the cowrie shells (Figure 3**).**

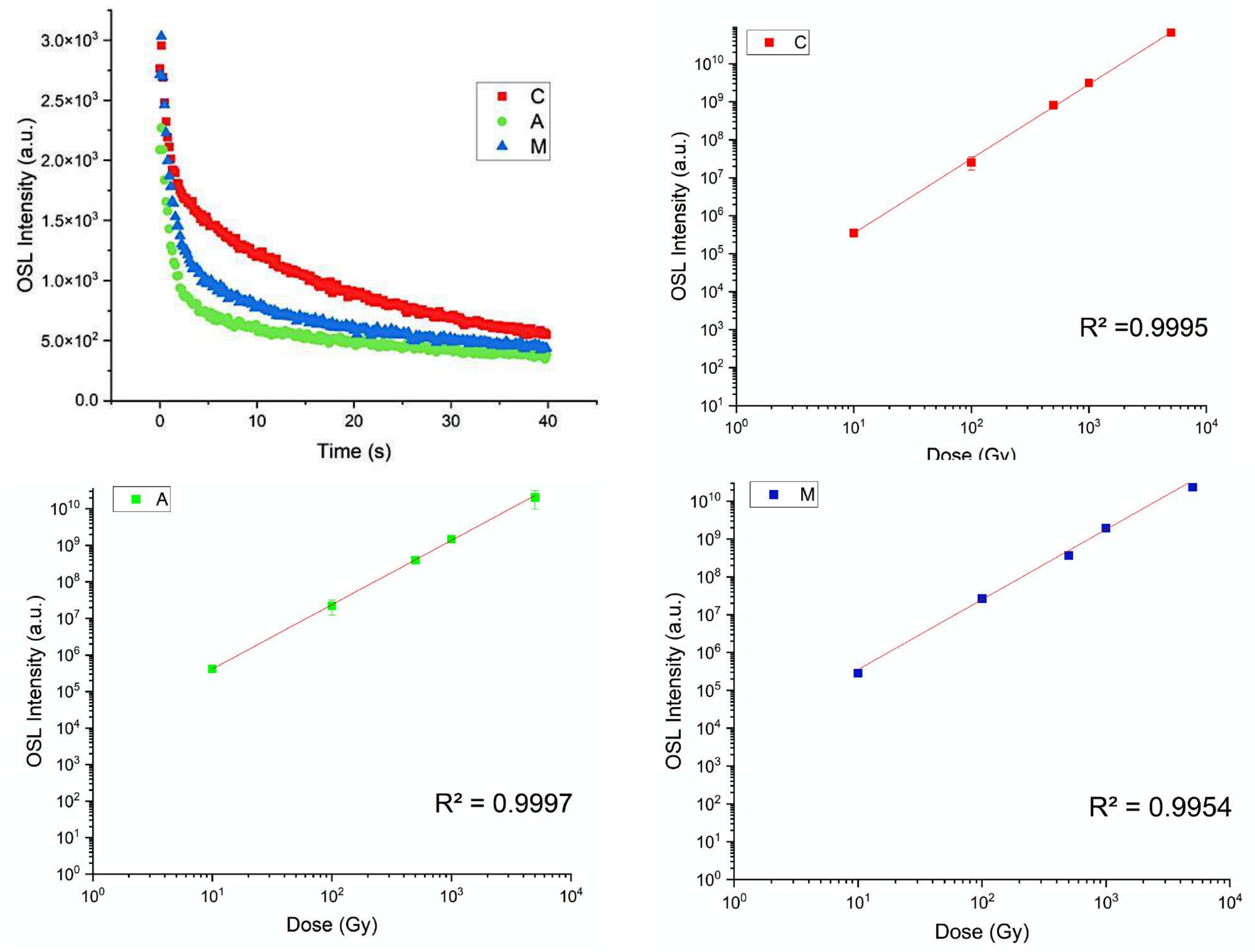

Figure 3: OSL decay curves and dose–response curves of the samples C, A and M shells in the range of 10 Gy – 5 kGy.

XRD measurements exhibit distinct crystal structures (Figure 4): Calcite in sample C, and Aragonite in samples A and M. These differences are related to natural environmental conditions and biological aspects of the shell-forming organisms.

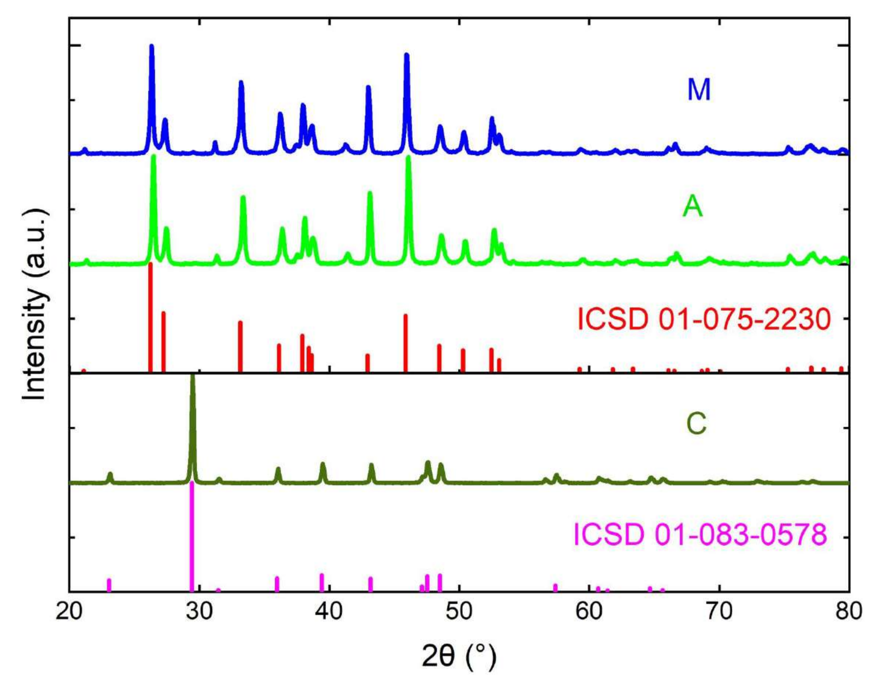

Figure 4: XRD patterns of calcined powders from shells C, A and M with corresponding crystallographic phases: Calcite (ICSD 01-083-0578) and Aragonite (ICSD 01-075-2230).

4. Conclusions

The use of cowrie shells biomass as a dosimetric material reinforces that they are a natural, low-cost, and environmentally friendly alternative. Their structural and luminescent properties support their application in radiation monitoring, particularly in industrial and environmental scenarios involving high radiation doses.

Reference

Ogundare, F. O., Chithambo, M. L., & Akintunde, B. O. (2021). Optically stimulated luminescence of cowrie shells. *Applied Radiation and Isotopes*, *167*, 109463.

Acknowledgements

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